

EEMB 120 -- Introduction to Ecology *Summer 2021, Session A*

Lecture:

- Weds, Thurs, Fri, 11:00am-12:25pm

Sections:

- Thurs. 1:00-2:20 pm
- Thurs. 3:00 – 4:20 pm
- Fri. 9:00 – 10:20 am

Instructor: Elizabeth Forbes

Usually in 1117 Noble Hall but for now, Zoom

elizabethforbes@ucsb.edu

Office Hours: Weds 12:30-1:30, or by appt

TA: Maggie Klope

mmklope@ucsb.edu

Office Hours: W-F by appt

Website:

<https://gauchospace.ucsb.edu/courses/course/view.php?id=83238>

Grading

Reading responses (weekly)	15 pts
Participation (see below)	5 pts
Exam 1: quantitative	25 pts
Exam 2: theory	25 pts
Population choice	10 pts
Annotated bibliography	10 pts
Final project	10 pts
Course total	100 pts

IMPORTANT Dates:

Exam 1: Wednesday, July 14th

Annotated bibliography portion of final project: Friday, July 16th (midnight)

Exam 2: Wednesday, July 28th

Finished Final Project: Friday, July 30th (midnight)

Optional Extra Credit: Wednesday, July 29th (submitted online)

ASSIGNMENTS:

- *Reading responses:* these will be due once per week before section (by Weds. at midnight). You will discuss the answers to these questions in section with Maggie. These questions will touch base with your understanding of the reading material (primary research articles) for that week!

- *Participation* will involve combinations of discussion, small breakout groups, in-class polls, and forums. If you are participating in section asynchronously (with Maggie's knowledge), please post 2 discussion questions/comments on your section's forum that week in lieu of 'in-person' participation.
- *Exam 1*: your first quiz/exam will focus on quantitative methods in ecology to predict population growth. It will consist of a combination of multiple, short-answer, and image-based questions. You will have a full day to take the 60-min exam online, but will have 90 minutes to complete it once you start.
- *Exam 2*: your second quiz/exam will focus on theory and concepts in community ecology and ecological organization on the local to global scale. It will consist of a combination of multiple, short-answer, and image-based questions. You will have a full day to take the 60-min exam online, but will have 90 minutes to complete it once you start.
- *Population choice*: you will choose the population for which you will be writing your annotated bibliography and synthesis paper (see below, and project description/grading rubric on Gauchospace).
- *Final Project*: your final project will be an annotated bibliography and synthesis response on a wildlife population of your choosing, that will address/describe that population at each level of ecological organization. The goal will be to apply the concepts you have learned this quarter in a synthetic way, and to learn more about a species that interests you; more details on the project description and rubric!
- *Extra credit (2pts)*: a set of five field observations: species-species and/or species-environment interactions (e.g. feeding interactions, competition interactions, parasitism interactions, etc.)
 - If possible, take a photo
 - Record date, time, prevailing weather
 - Describe the interaction in detail (3-5 sentences)
 - Include species names (use an online guidebook to look it up) in your descriptions
 - If you took a photo, include it!
 - Format: PPT, drawings, word doc, etc. (a field observation notebook is totally personalized!)

Reading:

Chapters are in Bowman, unless otherwise marked. **Bold is required reading.** (Parenthetical material is optional). See week by week schedule below for details! This is a living document and there will likely be changes; I will always update you if there are any changes, via email.

Attendance:

For any lecture or section that you will not be able to attend (including if you need to participate asynchronously this quarter for any reason), I ask that you provide me or Maggie reasonable notice ahead of time. You can 'miss' one section (and thus one section quiz) with no explanation needed, but please be sure to get in touch with Maggie if you run into any circumstances requiring you to miss more than one. We're operating on the honor system:

while I would generally take formal attendance in every class, these are extraordinary circumstances and some folks are going to be participating asynchronously.

I will be delivering each lecture 'live', but will also be recording each lecture and posting the video to Gauchospace ASAP after each class period. This way anyone who needs to participate asynchronously can do so. If you do plan to participate asynchronously, try to make sure you're not stuck in the position of having to review ~5+ hours of lecture on the weekend to catch up; there's a lot of material!

If you have concerns please do not hesitate to reach out to me (Elizabeth) to discuss them! Our goal is to optimize learning for the whole class.

How to access materials online:

All required (and some optional) materials will be posted online at our course Gauchospace. That's also where you'll find links to class documents, and the link to our Zoom meetings. I will be in touch regularly to update you on resources if they change!

Plagiarism:

Plagiarism is a serious subject. For all your work (in exams (written questions), weekly reading responses, and the final project), you will be expected and required to use your own words and interpretations of material; 'material' also encompasses lecture material/notes provided to you, as well as the work of your classmates. Any quotes must be properly attributed to their source, when used, though in this class it is always preferable to use your own phrasing whenever possible to demonstrate understanding of the material.

Code of conduct:

We will check in throughout our six weeks together, to make sure everybody's learning objectives are being met healthfully. Take this as your first of many invitations to speak to either of us (myself or Maggie) if you have any specific concerns. We encourage you to use the anonymous feedback form if you'd rather get in touch in that way. Our (metaphorical) doors are and will remain open.

Campus resources: *This has been a stressful year, and your mental health and safety are the most important part of any learning experience (including fast-paced summer courses like this one). Here are some opportunities for seeking aid in those arenas, all of which have been providing excellent support to students at UCSB this past year.*

- CARES: 24-hr advocacy line (805) 893-4613 [Zoom available],
- CAPS: new and initial appts scheduled at (805) 893-4411 [Zoom available]
- Health and Wellness: <http://wellness.sa.ucsb.edu>
- Student Health Center: telehealth, in person appointments, mailing of prescriptions <http://studenthealth.sa.ucsb.edu>

COURSE QUESTIONS:

Why do we care about learning about ecology?

What are the ways we're going to break it down, week by week?

How is ecology research conducted, and with what methods?

How does studying ecology get contextualized in/by our lives today?

LECTURE SYLLABUS:

Week of: **Topic:**

June 23	Introduction to Ecology & physical environment READINGS: Ch. 1 (2), this brief ESA article What Is Ecology? GUIDING QUESTIONS: • <i>What are we going to get out of this class?</i> • <i>What is ecological research, how is it done?</i>
June 24	Physiological Ecology: INDIVIDUAL READINGS: Ch. 4 (2,5), Williams et al 2014, Connor and Robles 2015 Dr. Kwasi M. Connor's website (UCI) Dr. Terrie Williams' lab website (UCSC) GUIDING QUESTIONS: • <i>How do organisms survive in their environment?</i> • <i>What kinds of limits does an environment put on species?</i>
June 25	Behavioral Ecology: INDIVIDUAL READINGS: Ch 8, Kienle et al. 2019 Meet Sarah Kienle , & advisor/co-author Dr. Rita Mehta (UCSC) GUIDING QUESTIONS: • <i>How do organisms adapt their behaviors in response to their environment?</i> • <i>What ecological factors drive changes in behavior?</i>
June 30	Population Ecology I- Exponential: POPULATION READINGS: Ch 9,10, Finkelstein et al 2012 , Gotelli 1 VIEWING: Khan Academy short video (watch this before class on Weds) Dr. Myra Finkelstein's website 2017 article on Dr. Finkelstein's research on condors
July 1	Population Ecology I – Exponential (continued): POPULATION GUIDING QUESTIONS: • <i>What do we mean when we refer to species population growth?</i> • <i>What are the factors that determine if a population is growing or not at any point in time?</i>
July 2	Population Ecology II – Age Structured growth: POPULATION READINGS: Ch 10, Fujiwara & Caswell 2001 , Gotelli 3 Dr. Masami Fujiwara's website at Texas A&M

GUIDING QUESTIONS:

- *How can we use the life stages and 'survivorship schedule' of a species to help us predict population growth?*

July 7	Population Ecology III - - Logistic growth: POPULATION READINGS: Ch 10-11, WNPR article on lobster fisheries , Gotelli 2 GUIDING QUESTIONS: • <i>What factors (species or environmental) might limit population growth?</i>
July 8	Species Interactions: Competition: COMMUNITY READINGS: Ch. 14 GUIDING QUESTIONS: • <i>How might competition for resources between species influence population growth of those species?</i>
July 9	Species Interactions: Predation: COMMUNITY READINGS: Ch. 12, Wilkinson et al. 2020 See Christine Wilkinson's website here, her profile on the Brashares lab website here , and her Twitter here GUIDING QUESTIONS: • <i>How does predation impact populations of both predators and prey?</i> • <i>How does predation impact species behavior?</i> NOTE: Final project population choice due by midnight today!

July 14	** Exam 1: quantitative topics in ecology, Weds July 14th**
July 15	Species Interactions: Parasitism: COMMUNITY READINGS: Ch. 13, Longo et al. 2020 , Gotelli 6 Check out Dr. Ana Longo, her lab group, and their research here ! GUIDING QUESTIONS: • <i>What is parasitism compared to other kinds of species relationships?</i> • <i>How do parasites impact populations, behaviors, and species interactions?</i>
July 16	Species Interactions: Complex interactions: COMMUNITY READINGS: Ch. 21 (16), Kefi et al. 2012 See Dr. Sonia Kefi's website here , as well as UCSB's and the UC Reserve system's own Dr. Carol Blanchette (co-author)! GUIDING QUESTIONS: • <i>What kinds of interactions can species have?</i> • <i>What kinds are feeding-based, and what kinds are non-feeding?</i>

**** Annotated bibliography due today, Fri. July 16th, by midnight ****

- July 21 Diversity 1 – Patterns of biodiversity: **ECOSYSTEM**
READINGS: Ch 18 and 19, **de Souza et al. 2020**, Gotelli 7,9
See Dr. Lesley de Souza's website [here](#), and give her a follow on [Twitter](#) or [Instagram](#)! (Dr. de Souza works at a museum, another cool way to be an ecologist.)
GUIDING QUESTIONS:
 - *How are species distributed and why?*
- July 22 Nutrient Cycling: **ECOSYSTEM** → **BIOSPHERE**
GUIDING QUESTIONS:
 - *What are nutrients, and why are they important?*
 - *How do nutrients cycle throughout an ecosystem, both the living and nonliving components?*
- July 23 Climate Change: **BIOSPHERE**
READINGS:
1: **Heat waves, wildfire, and permafrost thaw: The North's climate change trifecta** ([link here](#))
2: **Siberian heat streak and Arctic temperature record virtually 'impossible' without global warming, study says'** ([link here](#))
GUIDING QUESTIONS:
 - *How does a changing climate impact species, species interactions, populations, and distributions?*
 - *How does climate interact with questions of justice?*
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- July 28 **** Exam 2: theory and broad-scale concepts in ecology, Weds July 28th ****
- July 29 Conservation Ecology: **all the above!**
READINGS: Ch 23-24
1: **The concrete jungle has 578 miles of shoreline at risk** ([link here](#))
2: **Research breakthrough in fight against Chytrid fungus** ([link here](#))
GUIDING QUESTIONS:
 - *What is conservation ecology?*
 - *History of conservation*
 - *Based on what you know about species, interactions with each other and their environments, populations, and distributions, what are the strategies you'd favor for conservation, and why?*
 - *How does conservation interact with questions of justice?*

July 30

***** Final Project Due, midnight*****

Career options discussion and possible panel. Come prepared with questions about how to approach the many possible career options available in ecology!